



SOFTWARE REQUIREMENT SPECIFICATION (SRS)

ONLINE LIBRARY MANAGEMENT SYSTEM

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1. INTRODUCTION

1.1 Purpose

The purpose of the Online Library Management System (OLMS) is to transform traditional, manual library operations into a seamless, digital experience for both library staff and users. In today's fast-paced world, relying on paper-based records or outdated systems leads to inefficiencies—books go missing, tracking borrowings becomes chaotic, and users waste time visiting the library only to find their desired book unavailable. This system eliminates those frustrations by bringing the entire library online, making it accessible anytime, anywhere.

One of the biggest challenges in manual systems is the lack of real-time updates. Librarians struggle to keep track of borrowed books, leading to misplaced inventory and disputes. Users, on the other hand, have no way to check book availability without physically visiting the library. OLMS solves these problems by automating every critical process—book searches, reservations, checkouts, returns, and even late fee calculations—all in one centralized platform.

Beyond just convenience, this system significantly reduces the workload for library staff. Instead of manually recording transactions or handling overdue notices, librarians can focus on improving user experience, curating collections, or hosting library events. Automation also minimizes human errors, ensuring accurate record-keeping and smoother operations.

The system is designed with modern digital trends in mind. It's mobile-friendly, allowing users to browse and reserve books from their smartphones, and cloud-based, ensuring data is secure yet accessible from multiple devices. Whether it's a student checking book availability at midnight or a librarian generating reports during peak hours, OLMS adapts to real-world needs.

Problems with Manual Systems

1. **Lost or Misplaced Books:** Without real-time tracking, books often disappear from shelves, leaving users disappointed when they visit. Librarians waste hours searching for "available" books that are actually missing.

- *Example:* A student makes a 30-minute trip to borrow "Introduction to Algorithms," only to find it's been missing for weeks—despite the ledger showing it as "available."

2. **Inefficient Processes:**

- Staff manually record borrowings in logbooks, leading to errors (e.g., wrong due dates, misspelled names).
- Overdue fines are calculated by hand, causing disputes.
- Reservations are managed via phone calls or sticky notes, creating chaos.

3. **Limited Access:** Users can't check book availability outside library hours, and remote members (e.g., distance learners) are excluded.

How OLMS Solves These Problems

1. **24/7 Access:**

- Users browse the catalog, check availability, and reserve books via a mobile app or website—no physical visit needed.
- *Scenario:* A working professional reserves a book at midnight and picks it up the next day without asking staff for help.

2. **Real-Time Tracking:**

- Every book's status (available/borrowed/lost) updates instantly.
- Librarians scan books with barcode readers to log transactions in seconds.

3. **Automated Workflows:**

- The system auto-calculates fines based on due dates, sends reminders via email/SMS, and blocks users with unpaid fines.
- Generates reports (e.g., "Most Borrowed Books") with one click, saving hours of manual work.

4. **Inclusivity:**

- It supports multi-language interfaces and accessibility features (e.g., screen readers).
- Remote users enjoy equal access—no more "in-person only" policies.

By digitizing these processes, the library also becomes more sustainable—less paper waste, fewer physical logbooks, and reduced energy spent on repetitive tasks. Ultimately, this system isn't just about replacing pen-and-paper methods; it's about creating a smarter, more efficient library that serves its community better.

1.2 Document Conventions

The following conventions are used in the document:

Typography

Element	Formatting	Example
Main Headings (H1)	Times New Roman 16pt, Bold, Left-Aligned	1. Introduction
Sub-Headings (H2)	Times New Roman 14pt, Bold, Left-Aligned	1.1 Purpose
Labels/Tags (H3)	Times New Roman 12pt, Bold	FR-01: User login
Body Text	Times New Roman 12pt, Justified, 1.5 spacing	This is the standard paragraph.
Bullet Points	Times New Roman 12pt, Hyphen (-) or ▪	- Feature list

Table 1:document conventions

Requirement Labeling

- **Functional Requirements:** FR-XXX (e.g., FR-001: Book search)
- **Non-Functional Requirements:** NFR-XXX (e.g., NFR-003: Response time <2s)

Visuals Labeling

- **Tables:** Labeled as Table X (e.g., Table 1: Data Dictionary).
- **figures:** Numbered sequentially as Fig. X (e.g., Fig. 1: Use Case Diagram).

1.3 Project scope:

1. Business Requirements

1.1 Background

Libraries currently rely on manual processes for managing books, tracking inventory, handling member registrations, and processing borrow/return requests. This leads to inefficiencies, errors, and delays in operations. The **Online Library Management System (OLMS)** aims to automate these tasks, improving efficiency and user experience for both library staff and members.

1.2 Business Opportunity

The library management system presents a significant business opportunity by addressing three critical needs in modern library operations. First, there is growing demand for digital library solutions as institutions seek to modernize their services and meet patron expectations for online access. Second, libraries urgently require real-time book availability tracking and automated notification systems to improve service efficiency and user experience. Third, this solution offers libraries the opportunity to dramatically reduce administrative workload through automation while simultaneously enhancing member satisfaction with faster, more convenient services. Together, these factors create a compelling value proposition for libraries looking to transition from manual, paper-based processes to an efficient digital ecosystem that better serves both staff and patrons in today's technology-driven environment.

1.3 Business Objectives

- Turn hours of manual book tracking into a few clicks.
- Let borrowers reserve books online and get friendly reminders about due dates.
- Generate reports in seconds, not hours

- Ensure member information is protected like a treasured book.
- Automate book cataloging, borrowing, returning, and inventory management.
- Provide a user-friendly interface for members and staff.
- Enable real-time notifications for due dates, overdue books, and reservations.

1.4 Success Metrics

- The system will reduce book tracking mistakes by 80% compared to manual methods.
- Checking out and returning books will be 50% faster for both staff and members
- We aim for 90% of library members to be satisfied with the new system
- Real-time book availability updates should increase reservations by 30%
- Staff will spend less time fixing errors and more time helping members
- Members will spend less time waiting in line for basic services
- More books will get used as members can easily see what's available
- The library will operate more efficiently while providing better service

1.5 Vision Statement

"To revolutionize library operations by providing an automated, efficient, and user-friendly Online Library Management System that enhances accessibility, reduces manual effort, and improves member engagement."

1.6 Business Risks

- The notification system might take more time to develop than planned.
- Some librarians may prefer their familiar paper-based methods at first.
- Protecting user data will be extremely important for the system.
- Staff members used to manual processes may resist changing their routines.
- Keeping payment information and member details secure is a top priority.

- The system won't work properly without reliable internet access.
- Training staff on the new system may take longer than expected.
- Some members might need help learning to use the digital services.

1.7 Business Assumptions and Dependencies

- Libraries have internet (no dial-up allowed!).
- We'll partner with trusted tools for emails and SMS alerts.
- Library staff will be trained to use the system.
- Members will adopt the digital platform for borrowing/reservations.
- Integration with payment gateways for fine payments.

2. Scope and Limitations

2.1 Major Features

- **Book Management:** Add, update, and delete book metadata.
- **Member Management:** Register members and manage membership types (student, faculty, premium).
- **Borrowing/Returning:** Process checkouts, returns, and track due dates.
- **Fines and Payments:** Automatically calculate fines and process payments.
- **Reservations:** Allow members to reserve unavailable books.
- **Search Functionality:** Enable book searches by title, author, or genre.
- **Notifications:** Send alerts for due dates, overdue books, and reservations.

- **Reporting:** Generate reports on books, members, transactions, and fines.

2.2 Scope of Initial Release

- The system lets staff add, edit and remove book details like titles, authors and quantities.
- Users can register with their student ID or email and get different access levels.
- Members can borrow up to 5 books with automatic due date calculations.
- The system tracks returns and calculates late fines at \$0.50 per day.
- Librarians get a dashboard to manage checkouts, overdue items and reports.
- Members can search books, view their borrowing history and pay fines online.
- Automatic reminders go out before due dates and when books become available.
- The web interface works on computers and phones through modern browsers.

2.3 Scope of Subsequent Releases

- Users can search for books and make reservations through a mobile app.
- The app lets users scan barcodes to quickly find book information.
- Automatic notifications remind users when books are due for return.
- The system shows which books are most popular among members.
- Staff can view reports about member activity and borrowing patterns.
- Users receive personalized book suggestions based on their reading history.
- The system can display information in multiple languages.
- Members can save their digital library card to their phone.
- QR codes allow quick access to library services and accounts.
- The system sends alerts about new books in members' favorite categories.
- Notifications can be received by text message, email or chat apps.
- Members can schedule when they want to receive different types of alerts.

2.4 Limitations and Exclusions

Exclusions (What the System Will *Not* Do):

- **Physical Inventory Tracking**

Excluded: RFID/barcode scanning hardware integration (e.g., handheld scanners).

Reason: Scope is limited to digital management; physical tracking requires separate hardware budgets/vendors.

- **Offline Functionality**

Excluded: Operations without internet access (e.g., remote areas with poor connectivity).

Reason: Cloud-based design prioritizes real-time updates and centralized data.

- **Advanced Analytics**

Excluded: Predictive modeling (e.g., "Recommended for You" algorithms).

Reason: Phase 1 focuses on core transactions; analytics planned for future releases.

- **Inter-Library Loans**

Excluded: Integration with other libraries' catalogs.

Reason: Requires complex API partnerships beyond initial scope.

- **Multimedia Management**

Excluded: Handling DVDs, CDs, or other non-book materials.

Reason: Scope limited to traditional print/digital books.

- **Deep Customization**

Excluded: UI theme customization (e.g., library-branded color schemes).

Reason: Standardized interface for faster deployment.

Limitations (Constraints Within Scope)

- **Dependency on Third-Party Services**

SMS/email notifications rely on external providers (e.g., Twilio, SendGrid).

Impact: Service outages or API changes may temporarily disrupt notifications.

- **Data Migration Complexity**

Legacy manual records (paper logs, Excel files) require significant cleanup before importing.

Impact: Initial setup may take longer if historical data is inconsistent.

- **Browser Compatibility**

Fully optimized for modern browsers (Chrome, Firefox, Edge).

Limitation: Older browsers (e.g., Internet Explorer 11) are unsupported.

- **Mobile Responsiveness**

Web interface works on mobile browsers but lacks native app features (e.g., offline caching).

Impact: Mobile app development is deferred to Phase 2.

- **Concurrent User Load**

Supports up to 500 concurrent users during peak hours.

Limitation: May slow down during extreme spikes (e.g., exam periods).

- **Geographical Restrictions**

SMS notifications may not deliver reliably in all countries (carrier restrictions).

Workaround: Fallback to email notifications if SMS fails.

- **Payment Gateway Limitations**

Supports major credit cards but not regional payment methods (e.g., Meezan Bank mobile payments).

Reason: Integration complexity for local payment processors.

- **Accessibility Constraints**

Meets basic WCAG 2.1 standards but lacks advanced screen-reader customization.

Impact: Visually impaired users may need minor workarounds.

- **Data Export Flexibility**

Reports export to PDF/Excel but not real-time API integrations (e.g., Power BI).

Reason: Prioritized core functionality over advanced analytics.

- **Scalability**

Designed for single-library use; multi-branch scaling requires additional configuration.

3. Business Context

3.1 Stakeholder Profiles

Stakeholder	Role/Interest

Librarians	Manage books, process transactions, generate reports.
Members	Borrow books, track due dates, pay fines.
Administrators	Configure system settings, manage staff/membership types.
IT Team	Maintain system security and performance.

Table 2 : stakeholders

3.2 Project Priorities

1. High Priority

Essential for launch – impacts basic operations

- **Book Borrowing/Returning**
 - Checkout/check-in workflows with due-date tracking
 - Automatic fine calculation for overdue books
- **Real-Time Inventory Tracking**
 - Instant updates to book availability status
 - Prevention of double-booking conflicts
- **User Authentication & Roles**
 - Secure login for members/librarians
 - Role-based permissions (e.g., admin vs. student access)
- **Basic Search Functionality**
 - Title/author/ISBN search with filters
 - "Available Now" status indicators

2. Medium Priority

Improves usability but not critical for launch

- **Notifications System**

- Email/SMS alerts for due dates, reservations, and fines
- Configurable reminder schedules (e.g., 3-day warning)

- **Reporting Module**

- Predefined reports (overdue books, popular titles)
- Export to PDF/Excel (no real-time dashboards)

- **Reservation Management**

- Hold requests for checked-out items
- Auto-notification when reserved books are returned

- **Basic Fine Payment Integration**

- Credit card processing
- Receipt generation

3. Low Priority (Future Phase)

Post-launch optimizations

- **Mobile App Development**

- iOS/Android native apps with offline caching
- Push notifications (vs. SMS/email)

- **Advanced Analytics**

- Usage trends (peak hours, member demographics)

- Predictive "Recommended for You" features
- **Inter-Library Loans**
 - Integration with nearby libraries' catalogs
 - Cross-institution borrowing workflows
- **Accessibility Upgrades**
 - Screen-reader optimization beyond WCAG 2.1
 - Voice-command search
- **Hardware Integration**
 - RFID/barcode scanner support
 - Self-checkout kiosk compatibility

3.3 Deployment Considerations

- **Infrastructure & Hosting**
 - Cloud-Based Deployment**
 - Platform:** AWS/Azure (auto-scaling for peak loads)
 - Backup:** Daily encrypted backups with 30-day retention
 - Compliance:** GDPR/ISO 27001 for data security
 - Hybrid Option**
 - On-premise server for sensitive data (if required)
 - Cloud sync for public-facing services
- **Data Migration**
 - Source Data Handling**
 - Legacy formats:** Excel, CSV, or manual records

Data cleansing: Remove duplicates, standardize ISBNs

-Migration Process

Phase 1: Non-critical data (book catalog)

Phase 2: Member records (PII encrypted)

Validation: 100% manual review of 10% sample

-Fallback Plan

Parallel run: Old + new system for 2 weeks

Rollback window: 48 hours post-go-live

- **Training & Change Management**

-Stakeholder Groups

Group	Training Focus	Duration
Librarians	Transaction workflows, reports	8 hrs
IT Staff	System admin, troubleshooting	12 hrs
End-Users	Self-service portal orientation	1 hr (video)

Table 3

-Materials

Interactive sandbox environment for practice

QR-code linked video tutorials at workstations

- **Go-Live Strategy**

-Pilot Phase

1 test branch for 2 weeks

Feedback incorporated in 2 sprints

-Full Deployment

Weekend cutover with 24/7 support team

Real-time monitoring dashboard (uptime, errors)

- **Post-Deployment**

-Hypercare Period

Dedicated support team for 30 days

Priority ticketing for critical issues

-Performance Baseline

Key metrics:

Checkout time (<30 sec)

Search response (<2 sec)

Notification delivery rate (>99%)

- **Dependencies & Risks**

-Critical Dependencies

Stable internet (min. 10Mbps per branch)

Browser upgrades (Chrome v90+)

-Mitigation Plans

LTE failover for internet outages

Thin-client terminals as backup